

Chapter 1

Matrices and Linear Systems

1.1 Introduction to Matrices and Linear Systems

A system of m linear equations in the n unknowns $x_1, \dots, x_n$ has the form

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n &= b_3 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m\end{aligned}\tag{1.1}$$

is called a $(m \times n)$ **system of linear equations**, where the a_{ij} and b_i are fixed constants.

Example 1. Consider the following (3×4) linear system:

$$\begin{aligned}23x - 22y + 2w + 4z &= 26 \\3x + 2y - w + z &= 4 \\-10x - 5y + 4w - 7z &= -6.\end{aligned}\tag{1.2}$$

One can check that $x = 2, y = 1, w = 3, z = -1$ is a solution to this system. Thus, we will say that $(2, 1, 3, -1)$ is a solution to the system (1.2).

Definition 1.1.1. We say that the list of numbers $(x_1, \dots, x_n)$ is a solution to the system (1.1) if it satisfies **all** the equations in the system. The set

$$\mathcal{S} = \{(x_1, \dots, x_n) : (x_1, \dots, x_n) \text{ is a solution to the system (1.1)}\}$$

is called the **solution set** of the system (1.1). The system (1.1) is **consistent** if it has a solution, that is, $\mathcal{S} \neq \emptyset$. If the system (1.1) has no solutions then it is **inconsistent**.

Question. How to find the solution set $\mathcal{S}$ to the system (1.1)?

Answer. Transform the **complicated** system (1.1), using elementary operations (algebra), to an *equivalent* system which is **easier** to solve.

1.1.1 Elementary Operations on a Linear System of Equations

Definition 1.1.2. The **Elementary Operations** that can be applied to a system of equations are the following:

- (1) Multiply an equation by a non-zero number.
- (2) Interchange two equations.
- (3) Add a multiple of one equation to another equation.

Theorem 1.1.3. If one linear system of systems of equations is obtained from another linear system of equations by applying elementary operations, then both linear systems have exactly the same solutions.

Problem 2. Solve the system

$$\begin{aligned}x + 2y &= 8 \\ 3x - 4y &= 4.\end{aligned}$$

Solution. To be solved in class.

Problem 3. Solve the system

$$\begin{aligned}3x + 2y + z &= 2 \\ 4x + 2y + 2z &= 8 \\ x - y + z &= 4.\end{aligned}$$

Solution. To be solved in class.

1.1.2 Matrices

Definition 1.1.4. An $m \times n$ **matrix** is a rectangular array of numbers of the form

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

and the number a_{ij} is called the ij^{th} entry of the matrix A . We shall sometimes write $A = [a_{ij}]_{m \times n}$ as shorthand. Also, we shall sometimes write $A_{m \times n}$ when we want to emphasize the “size” of the matrix A . For notational convenience we shall write

$$\mathbf{a}_i = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}]$$

for the i^{th} row of the matrix A ; and we shall write

$$\mathbf{A}_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

for the j^{th} column of the matrix A .

For example, the 4×5 matrix

$$A_{4 \times 5} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ -2 & 0 & 20 & \pi & -4 \\ -3 & -1 & 3 & 6 & 9 \\ -4 & 4 & 5 & -4 & 8 \end{bmatrix}$$

has $a_{12} = 2$, and $a_{43} = 5$. The 2nd row of A is

$$\mathbf{a}_2 = \begin{bmatrix} -2 & 0 & 20 & \pi & -4 \end{bmatrix}$$

and the 3rd column of A is

$$\mathbf{A}_3 = \begin{bmatrix} 3 \\ 20 \\ 3 \\ 5 \end{bmatrix}.$$

1.1.3 Matrix Representation of a Linear System

In this section we will show how a system of equations can be expressed as a matrix. Matrices are helpful in rewriting a linear system in a very simple form. The algebraic properties of matrices may then be used to solve systems.

Given a system of m linear equations in the n unknowns $x_1, \dots, x_n$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \tag{1.3}$$

we form the two matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

The matrix A is called the **coefficient** matrix and the $m \times (n + 1)$ matrix

$$[A \mid \mathbf{b}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

is called the **augmented** matrix. The augmented matrix provides a more compact notation for a linear system of equations.

Example 4. Given the system of equations

$$\begin{aligned}x_1 + x_2 + 2x_3 &= 9 \\2x_1 + 9x_2 - 10x_3 &= 39 \\x_1 - 6x_2 + x_3 &= -27.\end{aligned}\tag{1.4}$$

the coefficient matrix is given by

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 9 & -10 \\ 1 & -6 & 1 \end{bmatrix}$$

and the matrix $\mathbf{b}$ is given by $\mathbf{b} = \begin{bmatrix} 9 \\ 39 \\ -27 \end{bmatrix}$. Thus, the augmented matrix of the linear system (1.4) is given by

$$[A \mid \mathbf{b}] = \begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 9 & -10 & 39 \\ 1 & -6 & 1 & -27 \end{bmatrix}.$$

1.1.4 Elementary Row Operations on a Matrix

We will now show how to use the augmented matrix to solve a system of equations.

Definition 1.1.5. The **Elementary Row Operations** which can be applied to a matrix are:

- (1) Multiply a row by a non-zero number
- (2) Interchange two rows
- (3) Add a multiple of one row to another.

Notation for Row Operations. Given a matrix A we shall let R_i represent the i -th row of A . Thus we can represent the above row operations as follows:

- (1) Replace the i -th row by a non-zero multiple k of the i -th row: $kR_i \rightarrow R_i$
- (2) Interchange the i -th and j -th rows: $R_i \leftrightarrow R_j$.
- (3) Replace the j -th row with a multiple of the i -th row added to the j -th row: $kR_i + R_j \rightarrow R_j$, where $i \neq j$.

Definition 1.1.6. If a matrix B can be obtained from a matrix A by performing a finite sequence of row operations on A , then the matrices A and B are said to be **row equivalent**.

Theorem 1.1.7. Let $[A \mid \mathbf{b}]$ and $[C \mid \mathbf{d}]$ be the augmented matrices of two linear systems of equations each of m equations and n unknowns. If the matrices $[A \mid \mathbf{b}]$ and $[C \mid \mathbf{d}]$ are row equivalent, then both linear systems have exactly the same solutions.

Problem 5. Find the solution set of the system of equations

$$\begin{aligned}x_1 + x_2 + 2x_3 &= 9 \\2x_1 + 9x_2 - 10x_3 &= 39 \\x_1 - 6x_2 + x_3 &= -27.\end{aligned}\tag{1.5}$$

Solution. We form the augmented matrix

$$[A \mid \mathbf{b}] = \begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 9 & -10 & 39 \\ 1 & -6 & 1 & -27 \end{bmatrix}.$$

Now, using row operations, we transform the matrix $[A \mid \mathbf{b}]$ into the “reduced echelon” form $[C \mid \mathbf{d}]$ as follows:

$$\begin{aligned} [A \mid \mathbf{b}] &= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 9 & -10 & 39 \\ 1 & -6 & 1 & -27 \end{bmatrix} \\ &\quad -2R_1 + R_2 \rightarrow R_2 \\ &= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 7 & -14 & 21 \\ 1 & -6 & 1 & -27 \end{bmatrix} \\ &\quad -R_1 + R_3 \rightarrow R_3 \\ &= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 7 & -14 & 21 \\ 0 & -7 & -1 & -36 \end{bmatrix} \\ &\quad R_2 + R_3 \rightarrow R_3 \\ &= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 7 & -14 & 21 \\ 0 & 0 & -15 & -15 \end{bmatrix} \\ &\quad \frac{1}{7}R_2 \rightarrow R_2 \\ &= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & -15 & -15 \end{bmatrix} \\ &\quad -\frac{1}{15}R_3 \rightarrow R_3 \\ &= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 1 & 1 \end{bmatrix} \\ &\quad -R_2 + R_1 \rightarrow R_1 \\ &= \begin{bmatrix} 1 & 0 & 4 & 6 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 1 & 1 \end{bmatrix} \\ &\quad 2R_3 + R_2 \rightarrow R_2 \\ &= \begin{bmatrix} 1 & 0 & 4 & 6 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 1 \end{bmatrix} \\ &\quad -4R_3 + R_1 \rightarrow R_1 \\ [C \mid \mathbf{d}] &= \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{aligned}$$

The matrix $[C \mid \mathbf{d}]$ is in “reduced echelon form”. We will now express this new augmented matrix $[C \mid \mathbf{d}]$ as a linear system of equations and solve this new system of equations to obtain:

$$\begin{aligned}x_1 &= 2 \\x_2 &= 5 \\x_3 &= 1\end{aligned}$$

This new system has the same solution set as (1.5) by Theorem 1.1.7 and thus, the solution set is $\mathcal{S} = \{(2, 5, 1)\}$.

Exercises 1.1

Pages 12 to 13 of text – #1, 3, 7, 9, 11, 13, 19, 21, 25, 27, 31, 35.

1.2 Reduced Echelon Form and Gauss-Jordan Elimination

Many of the problems we will solve in Linear Algebra require that a matrix be converted into *reduced echelon form*. Any matrix can be transformed to a matrix in reduced echelon form using elementary row operations, by a method known as Gauss-Jordan elimination. The solutions of a system of linear equations can be immediately obtained from the reduced echelon form to which the augmented matrix has been transformed.

Definition 1.2.1. A matrix is in **echelon form** if

- (a) All rows consisting of zeros, if any, are on the bottom of the matrix.
- (b) The first non-zero entry in each row is a 1, called the **leading one** of the row.
- (c) If two rows contain leading ones, the higher row will have its leading one to the left of the leading one of the lower row.

Definition 1.2.2. A matrix is in **reduced echelon form**¹

- (a) All rows consisting of zeros, if any, are on the bottom of the matrix.
- (b) The first non-zero entry in each row is a 1, called the **leading one** of the row.
- (c) If two rows contain leading ones, the higher row will have its leading one to the left of the leading one of the lower row.
- (d) A column containing a leading one has all other entries in the column equal to 0.

A matrix in reduced echelon form will have the general form

$$\begin{bmatrix} 1 & * & * & 0 & * & 0 & * & 0 & * & * & * \\ 0 & 0 & 0 & 1 & * & 0 & * & 0 & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & 0 & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where * represents any real number.

¹Some texts call this *reduced row echelon form*.

Example 1. Two matrices that are in reduced echelon form appear below:

$$M = \begin{bmatrix} 1 & 2 & 0 & 0 & 5 & -1 \\ 0 & 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad N = \begin{bmatrix} 1 & 2 & 3 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Theorem 1.2.3. Every matrix B is **row equivalent** to a matrix C in reduced echelon form.

We will illustrate the proof of Theorem 1.2.3 in our next example.

Example 2. Using elementary row operations we shall transform the matrix

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ -1 & 1 & 0 & -1 \end{bmatrix}.$$

into a matrix C in reduced echelon form as follows:

$$\begin{aligned} B &= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ -1 & 1 & 0 & -1 \end{bmatrix} \\ &\quad R_1 + R_3 \rightarrow R_3 \\ &\quad \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 3 & 3 & 3 \end{bmatrix} \\ &\quad -2R_2 + R_1 \rightarrow R_1 \\ &\quad \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 3 & 3 & 3 \end{bmatrix} \\ &\quad -3R_2 + R_3 \rightarrow R_3 \\ &\quad \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -3 & -6 \end{bmatrix} \\ &\quad -\frac{1}{3}R_3 \rightarrow R_3 \\ &\quad \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix} \\ &\quad -2R_3 + R_2 \rightarrow R_2 \\ &\quad \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \\ &\quad R_3 + R_1 \rightarrow R_1 \\ C &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \end{aligned}$$

The matrix C is in *reduced echelon form*. This completes the example.

1.2.1 Solving a Linear System by Reduction to Echelon Form

A system of m linear equations in the n unknowns $x_1, \dots, x_n$ has the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \tag{1.6}$$

where the a_{ij} and b_i are constants.

Definition 1.2.4. We say that the list of numbers $(x_1, \dots, x_n)$ is a solution to the system (1.6) if it satisfies **all** the equations in the system. The set

$$\mathcal{S} = \{(x_1, \dots, x_n) : (x_1, \dots, x_n) \text{ is a solution to the system (1.6)}\}$$

is called the **solution set** of the system (1.6). The system (1.6) is **consistent** if it has a solution, that is, $\mathcal{S} \neq \emptyset$. If the system (1.6) has no solutions then it is **inconsistent**.

Question. How to find the solution set $\mathcal{S}$ to the system (1.6)?

Answer. Since the algorithm we use to solve the system (1.6) makes no use of the variables $(x_1, \dots, x_n)$, we consider the augmented $m \times (n+1)$ matrix

$$[A \mid \mathbf{b}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

and perform *Row Operations* on $[A \mid \mathbf{b}]$ to transform this matrix to a new matrix $[C \mid \mathbf{d}]$ which is in *reduced echelon form*, obtaining the augmented matrix $[C \mid \mathbf{d}]$. The system of equations associated to the augmented matrix $[C \mid \mathbf{d}]$ is easy to solve and has exactly the same solutions as the original system (1.6).

1.2.2 Gauss-Jordan Elimination

We now give a procedure for solving a system of equations, by transforming the augmented matrix into another augmented matrix. The system represented by the new augmented matrix is easier to solve and has the same solution set as the original system of linear equations.

Gauss-Jordan Elimination Procedure. To solve a linear system of equations:

Step 1. Form the augmented matrix $[A \mid \mathbf{b}]$ of the system.

Step 2. Transform the augmented matrix $[A \mid \mathbf{b}]$ into *reduced echelon form* $[C \mid \mathbf{d}]$.

Step 3. Solve the new system of equations having the augmented matrix $[C \mid \mathbf{d}]$.

Problem 3. Find the solution set of the system of equations

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 4 \\x_2 + 2x_3 &= 3 \\-x_1 + x_2 &= -1.\end{aligned}\tag{1.7}$$

Solution. We apply Step 1 of the Gauss-Jordan elimination method, we obtain the augmented matrix

$$[A \mid \mathbf{b}] = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ -1 & 1 & 0 & -1 \end{bmatrix}.$$

We now apply Step 2 and transform this augmented matrix into the reduced echelon form $[C \mid \mathbf{d}]$. This was done in Example 2 above. Thus,

$$[C \mid \mathbf{d}] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Apply Step 3 and solve the the new system of equations with augmented matrix $[C \mid \mathbf{d}]$, having the same solution set as (1.7):

$$\begin{aligned}x_1 &= 0 \\x_2 &= -1 \\x_3 &= 2.\end{aligned}$$

and thus the solution set is $\mathcal{S} = \{(0, -1, 2)\}$.

Problem 4. Find the solution set of the system of equations

$$\begin{aligned}x_3 + 2x_4 - x_5 &= 4 \\x_4 - x_5 &= 3 \\x_3 + 3x_4 - 2x_5 &= 7 \\2x_1 + 4x_2 + x_3 + 7x_4 &= 7\end{aligned}\tag{1.8}$$

by Gauss-Jordan elimination.

Solution. We construct the the augmented matrix

$$[A \mid \mathbf{b}] = \begin{bmatrix} 0 & 0 & 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 3 & -2 & 7 \\ 2 & 4 & 1 & 7 & 0 & 7 \end{bmatrix}.$$

We now perform the following the row operations on the augmented matrix:

$$\begin{aligned}
 [A \mid \mathbf{b}] &= \begin{bmatrix} 0 & 0 & 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 3 & -2 & 7 \\ 2 & 4 & 1 & 7 & 0 & 7 \end{bmatrix} \\
 &\quad (R_1 \leftrightarrow R_4) \& (R_2 \leftrightarrow R_3) \\
 &\begin{bmatrix} 2 & 4 & 1 & 7 & 0 & 7 \\ 0 & 0 & 1 & 3 & -2 & 7 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 & -1 & 4 \end{bmatrix} \\
 &\quad -R_2 + R_4 \rightarrow R_4 \\
 &\begin{bmatrix} 2 & 4 & 1 & 7 & 0 & 7 \\ 0 & 0 & 1 & 3 & -2 & 7 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & -1 & 1 & -3 \end{bmatrix} \\
 &\quad R_3 + R_4 \rightarrow R_4 \\
 &\begin{bmatrix} 2 & 4 & 1 & 7 & 0 & 7 \\ 0 & 0 & 1 & 3 & -2 & 7 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\
 &\quad -R_2 + R_1 \rightarrow R_1 \\
 &\begin{bmatrix} 2 & 4 & 0 & 4 & 2 & 0 \\ 0 & 0 & 1 & 3 & -2 & 7 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\
 &\quad \frac{1}{2}R_1 \rightarrow R_1 \\
 &\begin{bmatrix} 1 & 2 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 3 & -2 & 7 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\
 &\quad (-3R_3 + R_2 \rightarrow R_2) \& (-2R_3 + R_1 \rightarrow R_1) \\
 [C \mid \mathbf{d}] &= \begin{bmatrix} 1 & 2 & 0 & 0 & 3 & -6 \\ 0 & 0 & 1 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

The matrix $[C \mid \mathbf{d}]$ is in reduced echelon form, and from $[C \mid \mathbf{d}]$ we get the following new system of equations which has the same solution set as (1.8):

$$\begin{aligned}
 x_1 + 2x_2 + \quad \quad \quad + 3x_5 &= -6 \\
 \quad \quad \quad x_3 \quad \quad + x_5 &= -2 \\
 \quad \quad \quad x_4 - x_5 &= 3.
 \end{aligned}$$

In the above system of equations, the variables x_1, x_3, x_4 are called the *leading variables* or the *dependent variables*. On the other hand, the variables x_2 and x_5 are called the *free variables* or the

independent variables. Now, solving for the leading variables x_1, x_3, x_4 we obtain

$$x_1 = -2x_2 - 3x_5 - 6$$

$$x_3 = -x_5 - 2$$

$$x_4 = x_5 + 3.$$

Since x_2 and x_5 are *free variables*, we will replace them with r and s , respectively, to obtain the solution

$$x_1 = -2r - 3s - 6$$

$$x_2 = r$$

$$x_3 = -s - 2$$

$$x_4 = s + 3$$

$$x_5 = s$$

where r and s are arbitrary, and hence, the solution set is

$$\mathcal{S} = \{(-2r - 3s - 6, r, -s - 2, s + 3, s) : r \text{ and } s \text{ are arbitrary real numbers}\}.$$

Therefore, there are an infinite number of solutions to the system (1.8).

Remark 1.2.5. When a consistent system has free variables in its “reduced echelon form system,” then the original system always has an infinite number of solutions.

1.2.3 How to Recognize an Inconsistent System?

Suppose we have the linear system of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \tag{1.9}$$

and we want to determine if the system has a solution. We form the augmented matrix

$$[A \mid \mathbf{b}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

and perform row operations on $[A \mid \mathbf{b}]$ to transform this matrix to a new matrix $[C \mid \mathbf{d}]$ in reduced echelon form. If $[C \mid \mathbf{d}]$ has the form, say

$$[C \mid \mathbf{d}] = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} & 0 \\ c_{21} & c_{22} & \cdots & c_{2n} & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix},$$

then the system (1.9) **has no** solutions. Why? Because this last non-zero row represents the equation

$$0x_1 + 0x_2 + \cdots + 0x_n = 1.$$

Since this equation has no solutions, the original system (1.9) is **inconsistent**, that is, the system (1.9) has no solutions.

Exercises 1.2

Pages 26 to 27 of text – #11, 13, 15, 17, 19, 23, 25, 27, 29, 37, 39.

1.3 Consistent Systems of Linear Equations

Theorem 1.3.1. Let $[A \mid \mathbf{b}]$ be the augmented matrix for the *consistent* system in n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned} \tag{1.10}$$

Let $[C \mid \mathbf{d}]$ be the reduced echelon form of $[A \mid \mathbf{b}]$. Suppose that $[C \mid \mathbf{d}]$ has r many leading ones. Then $r \leq n$ and there are $n - r$ free variables in the final solution to (1.10). In addition,

1. If $r = n$, then the solution to (1.10) is unique.
2. If $r < n$, then there are an infinite number of solutions to (1.10).

Proof. (Review Problem 4 on page 20.) We consider the augmented matrix

$$[A \mid \mathbf{b}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

and perform row operations on $[A \mid \mathbf{b}]$ to transform this matrix to the new matrix $[C \mid \mathbf{d}]$ in reduced echelon form. Since the system (#) is consistent, the reduced echelon matrix $[C \mid \mathbf{d}]$ must be of the form (with all non-zero rows on top)

$$[C \mid \mathbf{d}] = \begin{bmatrix} 1 & c_{12} & c_{13} & \cdots & c_{1n} & d_1 \\ 0 & c_{22} & c_{23} & \cdots & c_{2n} & d_2 \\ 0 & 0 & c_{33} & \cdots & c_{3n} & d_3 \\ 0 & 0 & 0 & \cdots & c_{4n} & d_4 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & c_{rn} & d_r \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}.$$

Let r be the number of leading ones in the matrix $[C \mid \mathbf{d}]$. Recall that n is the number of variables in the system $(\#)$ and that n is also the number of columns in the matrix C . Each non-zero row in $[C \mid \mathbf{d}]$ has exactly one leading 1. Since the system is consistent, we see that no leading 1 can occur in $\mathbf{d}$. So all of the leading 1's must occur in some (or all) of the columns in C . Therefore, $r \leq n$.

In addition, if $r = n$, then there is a unique solution, because there would be no free variables. Also, if $r < n$, then there are $n - r > 0$ many free variables and hence, there are an infinite number of solutions (we may choose **any values** we want for the free variables and then get the required values for the leading variables and thus, we can get an infinite number of solutions to the system). This completes the proof. $\square$

Corollary 1.3.2. Let $[A \mid \mathbf{b}]$ be the augmented matrix for the system in n unknowns and m equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned} \tag{1.11}$$

If $m < n$, then either the system (1.11) is inconsistent or it has an infinite number of solutions.

Proof. Assume that $m < n$. If the system is inconsistent, then we are done. So, suppose that the system is consistent. Clearly, the number of rows in the reduced echelon form $[C \mid \mathbf{d}]$ is equal to m (since the matrix $[A \mid \mathbf{b}]$ has m rows). Hence, one can see that the number r of non-zero rows in the matrix $[C \mid \mathbf{d}]$ must satisfy $r \leq m$. Now since $m < n$, it follows that $r < n$. Theorem 1.3.1 implies (see item 2 above) that the system must have an infinite number of solutions. $\square$

1.3.1 Homogeneous Linear Systems of Equations

Definition 1.3.3. A system of linear equations is said to be *homogeneous* if all the b_i 's are 0, that is, the system has the following form:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0. \end{aligned} \tag{1.12}$$

Remark 1.3.4. The homogeneous system (1.12) is *always consistent* because it has the **trivial solution** $x_1 = 0, x_2 = 0, \dots, x_n = 0$. We are interested in homogeneous systems which have **non-trivial solutions**.

The next Theorem 1.3.5 gives a condition which, when satisfied, will guarantee that a homogeneous system has an infinite number of solutions. This theorem follows immediately from the above Corollary 1.3.2; however, we shall present a proof of Theorem 1.3.5 that “illustrates” the proof of Corollary 1.3.2.

Theorem 1.3.5. If the homogeneous system (1.12) of linear equations has more unknowns than equations (that is, if $m < n$), then the system has infinitely many solutions.

Proof. Since the argument is easy to generalize, we shall consider only the special case when $m = 3$ and $n = 5$. So, let $[A \mid \theta]$ be the augmented matrix

$$[A \mid \theta] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & 0 \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} & 0 \end{bmatrix}.$$

After performing row operations on $[A \mid \theta]$, we can transform this matrix to a new matrix $[C \mid \theta]$ in reduced echelon form. [**Note: the column of zeros θ is preserved under elementary row operations.**] Now, because there are only three rows in the matrix $[A \mid \theta]$, there can be at most three leading 1's in the matrix $[C \mid \theta]$. (This implies that at least two of the five unknowns can be chosen arbitrarily.) Without loss of generality we can assume that $[C \mid \theta]$ has the form

$$[C \mid \theta] = \begin{bmatrix} 1 & c_{12} & 0 & 0 & c_{15} & 0 \\ 0 & 0 & 1 & 0 & c_{25} & 0 \\ 0 & 0 & 0 & 1 & c_{35} & 0 \end{bmatrix}$$

Since x_1, x_3, x_4 are leading variables, x_2 and x_5 are free variables. We will replace x_2 and x_5 with r and s , respectively, to obtain the system solution by solving for the leading variables x_1, x_3, x_4 (see the previous examples):

$$\begin{aligned} x_1 &= -c_{12}r - c_{15}s \\ x_2 &= r \\ x_3 &= -c_{25}s \\ x_4 &= -c_{35}s \\ x_5 &= s \end{aligned}$$

where r and s are arbitrary, and hence, there are an infinite number of solutions to the system. This completes the proof of the theorem. $\square$

Exercises 1.3

Pages 37 to 38 of text – #1, 2, 3, 4, 21, 22, 23, 24.

1.5 Matrix Operations

Definition 1.5.1. An $m \times n$ **matrix** is a rectangular array of numbers of the form

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

and the number a_{ij} is called the ij^{th} entry of the matrix A . We shall sometimes write $A = [a_{ij}]_{m \times n}$ as shorthand. Also, we shall sometimes write $A_{m \times n}$ when we want to emphasize the “size” of the matrix A . For notational convenience we shall write

$$\mathbf{a}_i = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}$$

for the i^{th} row of the matrix A ; and we shall write

$$\mathbf{A}_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

for the j^{th} column of the matrix A .

For example, the 4×5 matrix

$$A_{4 \times 5} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ -2 & 0 & 20 & \pi & -4 \\ -3 & -1 & 3 & 6 & 9 \\ -4 & 4 & 5 & -4 & 8 \end{bmatrix}$$

has $a_{12} = 2$, and $a_{43} = 5$. The 2nd row of A is

$$\mathbf{a}_2 = \begin{bmatrix} -2 & 0 & 20 & \pi & -4 \end{bmatrix}$$

and the 3rd column of A is

$$\mathbf{A}_3 = \begin{bmatrix} 3 \\ 20 \\ 3 \\ 5 \end{bmatrix}.$$

Definition 1.5.2. Two matrices, say $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$, are equal if and only if they have the same size and entries, that is, $a_{ij} = b_{ij}$ for all i, j .

1.5.1 Matrix Addition and Scalar Multiplication

Definition 1.5.3. Given $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$, we define $A + B = [c_{ij}]_{m \times n}$ where $c_{ij} = a_{ij} + b_{ij}$.

Example 1.

$$\begin{bmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & 0 & -40 \\ 2 & 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 & 3 \\ 1 & 0 & 1 & 41 \\ 4 & 0 & -1 & 9 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 1 & 6 \\ 1 & 1 & 1 & 1 \\ 6 & 0 & 2 & 10 \end{bmatrix}$$

Definition 1.5.4. Given $A = [a_{ij}]_{m \times n}$ and a scalar (real number) d , define **scalar multiplication** by $dA = [da_{ij}]_{m \times n}$.

Example 2.

$$2 \begin{bmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & 0 & -40 \\ 2 & 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 & 4 & 6 \\ 0 & 2 & 0 & -80 \\ 4 & 0 & 6 & 2 \end{bmatrix}$$

Determinants: An Overview

For each $n \times n$ matrix, A , we can calculate a number called the determinant of A , $\det(A)$. This is often written as $|A|$.

Case 1. 1×1 matrices. If $A = [a]$ then $\det(A) = |A| = a$.

Case 2. 2×2 matrices. Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, then we have:

$$\det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}.$$

Ex. $\begin{vmatrix} 3 & -2 \\ 4 & 5 \end{vmatrix} = 3(5) - (4)(-2) = 15 + 8 = 23.$

Def. Let $A = [a_{ij}]$ be an $n \times n$ matrix and let M_{ij} be the $(n-1) \times (n-1)$ matrix obtained from deleting the row and column containing a_{ij} . The determinant of (M_{ij}) is called the **Minor** of a_{ij} . We define the **Cofactor**, A_{ij} , of a_{ij} by:

$$A_{ij} = (-1)^{i+j} \det(M_{ij}).$$

Def. The **Determinant of an $n \times n$ matrix A** , where $A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$, is given

by: $\det(A) = a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n}$

where $A_{1j} = (-1)^{1+j} \det(M_{1j})$; $j = 1, 2, \dots, n$.

Case 3. 3x3 matrices. Let A be:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Then $\det(A) = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$;

where: $A_{11} = (-1)^{1+1} \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} = \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$

$$A_{12} = (-1)^{1+2} \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} = -\det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}$$

$$A_{13} = (-1)^{1+3} \det \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \det \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}.$$

Ex. Find $\det(A)$ where $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$.

$$\begin{vmatrix} 2 & 1 & 3 \\ 4 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} - 1 \begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix} + 3 \begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix}$$

$$= 2[(1)(3) - 2(2)] - 1[4(3) - 1(2)] + 3[4(2) - 1(1)]$$

$$= 2(3 - 4) - 1(12 - 2) + 3(8 - 1)$$

$$= 2(-1) - 10 + 3(7)$$

$$= 9.$$